LipScribe presenter notes

Technical implementation presentation | CVDL Group 4

These notes go with the 5-minute deck *LipScribe_Implementation_Presentation_5min.pptx*: two content slides per presenter, about 22 seconds each. For every slide they give the script, the background you need to explain it without reading, and questions you may be asked. The background is deliberately detailed: the slides are short, but questions can go into any detail of the implementation plan. The script is also in each slide's speaker notes.

Read the whole document, not only your own part. Any member may be questioned on any part of the project.

Who presents what

Part	Topic	Presenter	Slides	Time
1	Problem and architecture	Cole Rodney Migisha	1–3	0:59
2	Data, split and noise	Charis Opol	4–5	0:44
3	Visual and audio pipelines	Ainomugisha Faith	6–7	0:44
4	Fusion, dropout and CTC	Lisa Lusinga Namugenyi	8–9	0:44
5	Baselines and evaluation	Ansasire Nehum Naamaye	10–11	0:44
6	Demo, plan and risks	Wakisa Isabelle	12–16	0:54
	Total		1–16	4:49

Cole also opens the talk (title slide). Isabelle closes it (thank you) and chairs questions. Reference slides are backup only.

Timing

The whole talk is 5 minutes: 15 seconds of opening, two slides of about 22 seconds per presenter (45 seconds each), and 10 seconds to close. Each slide's script is about 40 to 55 words, which is about 22 seconds at a normal speaking pace. Point at the diagram as you speak; the slide carries the detail, the script carries the message. Rehearse with a timer; if you run long, drop the last sentence before your handover line, never the handover itself. The two reference slides come after the thank-you slide as backup and are not presented. The full 61-slide deck can be used as a backup to show detail during questions.

The one-sentence version

Captioning tools that only listen fail in noise. LipScribe also reads the lips, and learns through cross-modal attention and modality dropout to lean on them when the audio degrades. We will measure this on held-out GRID speakers at five noise levels against three baselines.

Agree these as a group before you present

- **There are no results yet.** This is the implementation plan. The results table is deliberately empty. Never quote a WER as achieved.
- **Targets are hypotheses:** WER below 10% on clean GRID audio, and a clear advantage over audio-only at 0 and -5 dB.

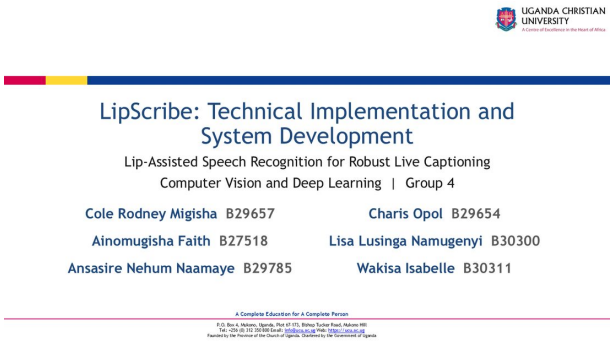
- **Data:** GRID, 34 speakers x 1,000 sentences, about 33,000 usable, 51-word vocabulary, 25 fps, about 75 frames per 3 s clip.
- **Noise:** MUSAN noise subset at clean, 10, 5, 0 and -5 dB SNR, with disjoint noise-file pools for train, validation and test.
- **Split:** by speaker. About 27 train, 3 validation, 4 test (2 female, 2 male). Never by sentence.
- **Shapes:** mouth crops 96 x 96 grey; both encoders output B x 75 x 512; fusion width 256; 28 output symbols (blank, space, a-z).
- **Labels in the document:** [Proposal] = fixed by the proposal; [Recommended] = our engineering choice, may change; [To be decided] = settled by experiment.
- **Handover line:** each presenter ends with one sentence naming the next presenter and their topic. These lines are already in the scripts.

Part 1: Problem and architecture

Presenter: **Cole Rodney Migisha**

Slides 1–3 | suggested total 0:59

Slide 1 | LipScribe: Technical Implementation (0:15)



WHAT TO SAY

- Good afternoon. We are Group 4. This is LipScribe, and today we present how we will build it. Each of us presents two slides.

BACKGROUND YOU SHOULD UNDERSTAND

This presentation covers the Technical Implementation and System Development Document. It describes planned work: there are no experimental results yet. Every number not fixed by the proposal is a recommendation or a target.

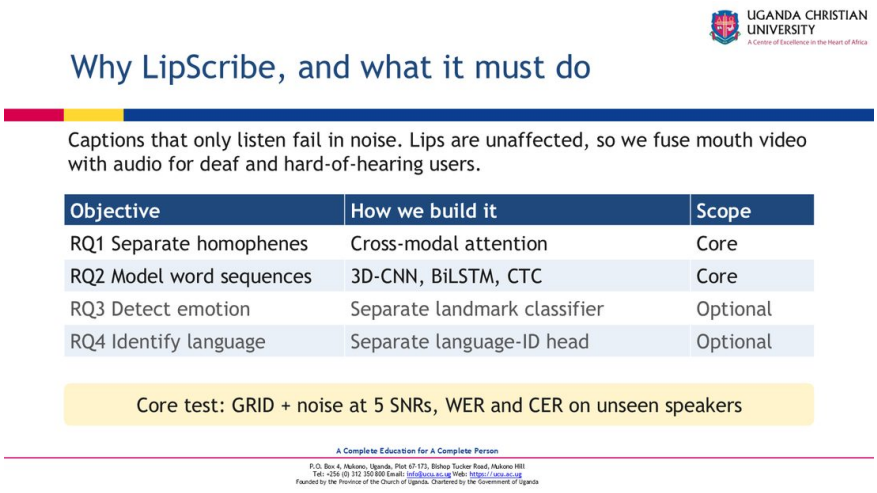
Statement labels in the document: [Proposal] = fixed by the proposal; [Recommended] = our engineering choice, may change; [To be decided] = settled by experiment.

QUESTIONS YOU MAY BE ASKED

Do you have results yet?

No. This is the implementation plan. Results come from logged training runs during the six weeks.

Slide 2 | Why LipScribe, and what it must do (0:22)



WHAT TO SAY

- Captioning tools that only listen fail in noise. Lips are not affected by noise, so LipScribe fuses mouth video with audio, for deaf and hard-of-hearing users.

- Our core objectives are separating homophenes and modelling word sequences. Emotion and language identification are optional, separate modules. We test on GRID with noise at five levels, on unseen speakers.

BACKGROUND YOU SHOULD UNDERSTAND

Objectives: (1) separate homophenes, sounds that look identical on the lips such as /p/, /b/, /m/; cross-modal attention lets audio resolve them. (2) Model word sequences: 3D-CNN, BiLSTM encoders and CTC map frames to sentences. (3) Emotion and (4) language identification need data GRID does not have (no emotion labels, one language), so they are separate optional modules started only after the core freeze.

WER and CER are word and character error rates (Nehum's slides). Five SNR levels: clean, 10, 5, 0 and -5 dB.

QUESTIONS YOU MAY BE ASKED

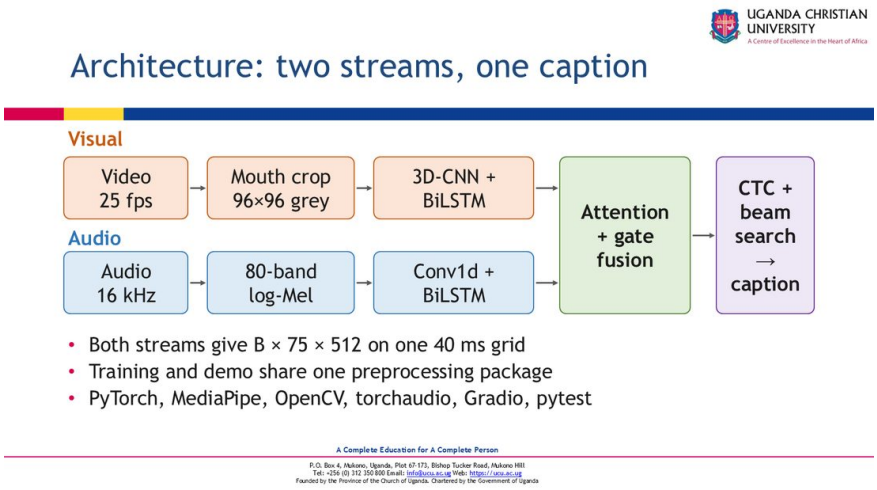
Why is this computer vision and not speech recognition?

The information that survives noise is visual. The new work is locating the mouth in video, encoding its shape and motion over time, and fusing it with audio. Audio-only recognition is our baseline to beat.

Why not include emotion detection in the main model?

GRID has no emotion labels, so it cannot be trained or evaluated there. Keeping it separate stops an unvalidated extension from delaying the main result.

Slide 3 | Architecture: two streams, one caption (0:22)



WHAT TO SAY

- Two streams. Visual: a 96 by 96 mouth crop into a 3D-CNN and BiLSTM. Audio: 80-band log-Mel features into a 1D convolution and BiLSTM.
- Attention and a gate fuse them, and CTC with beam search produces the caption. Both streams share one 40 millisecond grid. Charis will explain the data.

BACKGROUND YOU SHOULD UNDERSTAND

Eight run-time stages: capture audio and video; detect the face with MediaPipe Face Mesh; crop the mouth; extract log-Mel features; encode each stream; fuse with cross-modal attention; decode with CTC and beam search; display the caption.

Tools: Python 3.11, PyTorch (Conv3d, BiLSTM, multi-head attention and CTC loss built in), torchaudio for features, OpenCV and MediaPipe for video, Gradio for the web demo, jiwer for error rates, pytest for tests.

Training and inference share one preprocessing package, and a parity test checks that a GRID clip processed by the live path matches the cached training features.

QUESTIONS YOU MAY BE ASKED

What is a BiLSTM?

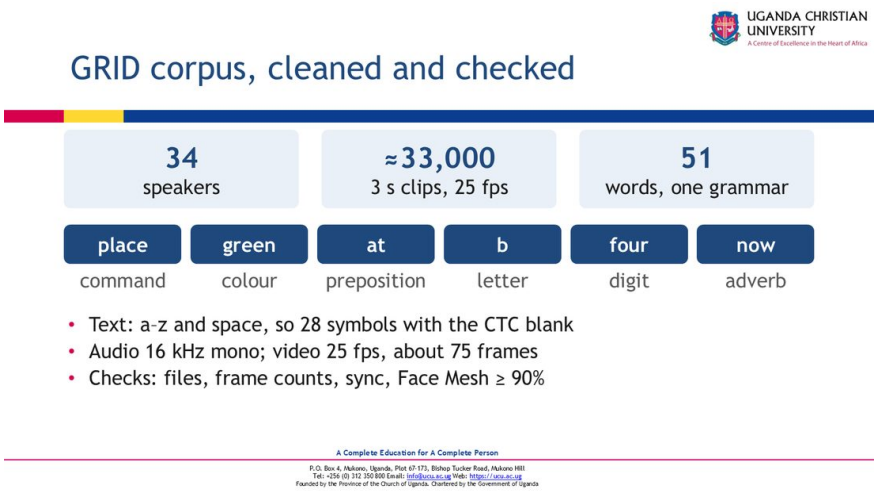
A recurrent network that reads the sequence forwards and backwards, so each frame's output uses past and future context.

Part 2: Data, split and noise

Presenter: **Charis Opol**

Slides 4–5 | suggested total 0:44

Slide 4 | GRID corpus, cleaned and checked (0:22)



WHAT TO SAY

- We use the GRID corpus: 34 speakers, about 33,000 three-second clips, 51 words in one six-slot grammar, like 'place green at b four now'.
- Text becomes 28 symbols, audio is 16 kHz, video 25 frames per second, and a script checks files, frame counts, sync and face detection before training.

BACKGROUND YOU SHOULD UNDERSTAND

Integrity checks (verify_dataset.py): all files present; video has 75 +/- 2 frames; audio and video lengths match within 40 ms; six valid GRID words; Face Mesh works on at least 90% of frames; audio not silent or clipped.

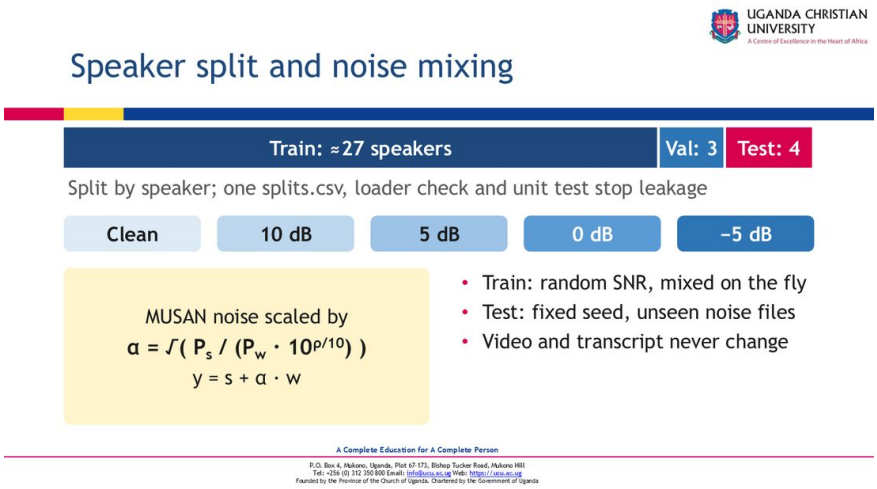
Text: lower-case a-z and space, digits as words, giving 28 output symbols including the CTC blank. Hardware: a GPU of about 8 GB for batch 16 with mixed precision; about 23 GB of cached crops; about 100 GB free disk recommended.

QUESTIONS YOU MAY BE ASKED

Isn't GRID too easy?

Yes, compared with real captioning, and we say so. It gives a controlled setting in which we can measure the effect of noise and of adding video cleanly. Larger datasets are future work.

Slide 5 | Speaker split and noise mixing (0:22)



WHAT TO SAY

- We split by speaker: about 27 to train, 3 to validate, 4 to test, so the model cannot memorise test faces.
- MUSAN noise is scaled by alpha to hit an exact SNR. In training the level is random; for testing it is fixed, with unseen noise files. Faith takes the pipelines.

BACKGROUND YOU SHOULD UNDERSTAND

Split safeguards: splits.csv is the only source of membership, the loader rejects a speaker found in two partitions, and a unit test asserts the sets are disjoint. The test set has 2 female and 2 male speakers. Normalisation statistics come from training speakers only.

SNR (signal-to-noise ratio): 10 dB means speech is 10 times more powerful than noise, 0 dB equal, -5 dB noise about three times stronger. Noise is scaled by alpha = sqrt(Ps / (Pw x 10^(rho/10))) so the SNR is exact, and the code checks it to within 0.01 dB. If the mixture's peak exceeds 1, the whole mixture is scaled down, which keeps the SNR.

Training: an SNR is drawn per clip with equal probability and noise is mixed on the fly. Validation and test: each clip is mixed once per SNR with a fixed seed and stored, so every model sees identical audio. Noise changes only the audio; video and transcript stay the same, so the same sentences are scored at every SNR.

QUESTIONS YOU MAY BE ASKED

With only 4 test speakers, are results reliable?

Limited. We report per-speaker WER, use up to three seeds, and use a paired bootstrap for a confidence interval on the main comparison.

Why use different noise files for testing?

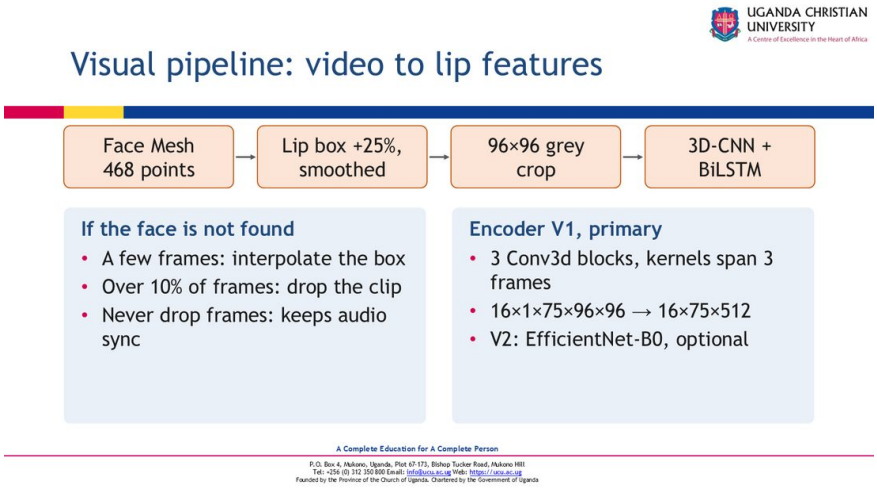
So we measure robustness to new noise, not recognition of noise the model memorised.

Part 3: Visual and audio pipelines

Presenter: **Ainomugisha Faith**

Slides 6–7 | suggested total 0:44

Slide 6 | Visual pipeline: video to lip features (0:22)



WHAT TO SAY

- MediaPipe Face Mesh finds 468 landmarks. We box the lips with a 25% margin, smooth it, and crop 96 by 96 grey images.
- Missing faces are interpolated, never dropped, to keep sync. A 3D-CNN with BiLSTM turns each clip into 75 vectors of 512. EfficientNet-B0 is an optional transfer-learning variant.

BACKGROUND YOU SHOULD UNDERSTAND

Mouth crop details: lip landmarks come from MediaPipe's FACEMESH_LIPS list; box side L = (1 + 2m) x S with m = 0.25; centre and size smoothed with a 5-frame moving average offline (exponential moving average in the live demo). 96 x 96 is about 27% smaller than 112 x 112 and is comparable to crop sizes used for GRID lipreading [12]. Grey-scale removes colour differences between speakers.

Failures: a few failed frames get an interpolated box and are marked invalid; clips with over 10% failures are excluded; lengths are padded or trimmed to the audio length.

Visual encoder V1 (primary): three Conv3d blocks (32/64/96 channels) whose kernels span 3 frames, so they capture motion such as lips closing; temporal stride 1 keeps one vector per frame; flatten 96x6x6 = 3456, linear to 512, 2-layer BiLSTM (256 per direction). V2 (optional ablation): a 3D stem then an ImageNet-pretrained EfficientNet-B0 per frame, our transfer-learning experiment.

QUESTIONS YOU MAY BE ASKED

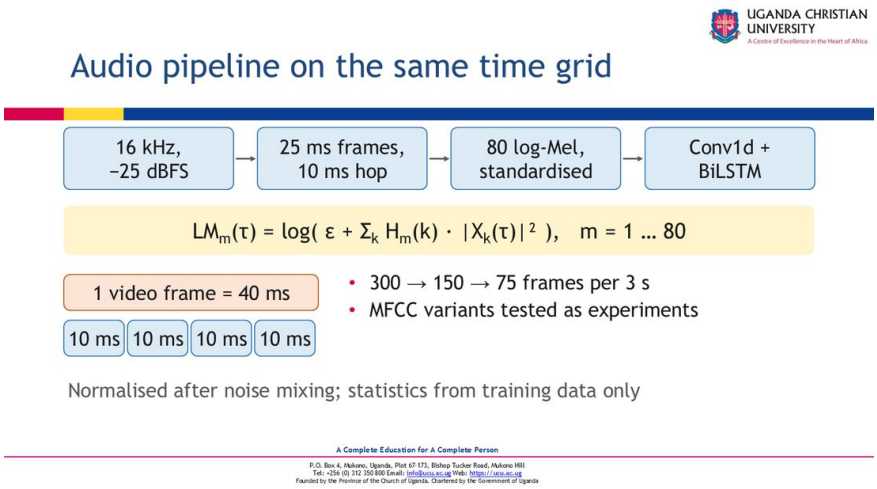
Why not drop frames where no face is found?

Dropping frames would break the fixed four-to-one alignment with audio. Interpolating keeps both streams on the same time grid.

What does a 3D convolution add over 2D?

Its kernel also slides over time, so one feature can respond to motion across consecutive frames, which a single image cannot show. Ablation X3 tests this.

Slide 7 | Audio pipeline on the same time grid (0:22)



WHAT TO SAY

- Audio is normalised, cut into 25 millisecond frames every 10 milliseconds, and turned into 80 log-Mel bands.
- One video frame equals four audio frames, so the encoder reduces 300 frames to 75 and both streams line up. MFCC variants are experiments. Lisa will explain the model.

BACKGROUND YOU SHOULD UNDERSTAND

Audio: RMS level -25 dBFS with gain capped at +20 dB; normalisation after noise mixing so SNR is unchanged; 512-point STFT; 80 Mel filters from 20 Hz to 8 kHz; log. 300 feature frames per 3 s -> two stride-2 convolutions -> 75 frames. MFCC (39) and log-Mel + MFCC (119) are tested as experiments on the audio-only baseline.

Tests: crop shape and type; log-Mel matches librosa within 1e-3; 3 s gives exactly 300 then 75 frames; both encoders output B x 75 x 512.

QUESTIONS YOU MAY BE ASKED

Why compute audio features on the fly but cache mouth crops?

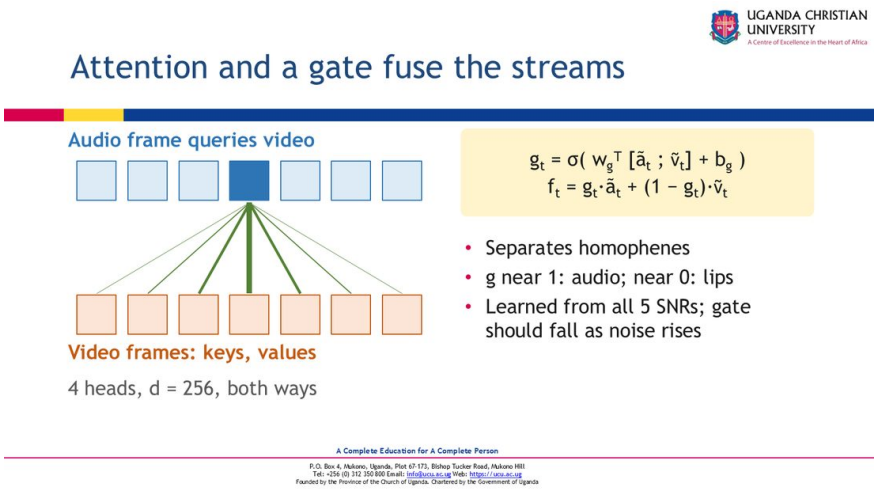
Noise changes every time a clip is loaded during training, so the audio features change; the mouth crops never do.

Part 4: Fusion, dropout and CTC

Presenter: **Lisa Lusinga Namugenyi**

Slides 8-9 | suggested total 0:44

Slide 8 | Attention and a gate fuse the streams (0:22)



WHAT TO SAY

- Cross-modal attention lets each audio frame look at the video frames, and the reverse, so each stream resolves the other's ambiguities, like homophenes.
- A learned gate then mixes them per frame: near one trusts audio, near zero trusts the lips. Trained on all noise levels, it should lean on lips as noise rises.

BACKGROUND YOU SHOULD UNDERSTAND

Model sizes: each encoder outputs B x T x 512; each is projected to 256 with LayerNorm; cross-attention runs both ways with 4 heads of 64; gate g_t = sigmoid(w . [a_t ; v_t] + b) in (0, 1); then a 1-layer BiLSTM (128 per direction) and a linear layer to 28 symbols with log-softmax.

Noise adaptation: attention does not detect noise by itself. We use implicit learning, training on all five SNRs plus modality dropout. Feeding the true SNR is impossible live; a learned SNR-estimation head is optional experiment X6. Check: the mean gate value should fall as SNR falls, and test-time ablations give direct evidence.

QUESTIONS YOU MAY BE ASKED

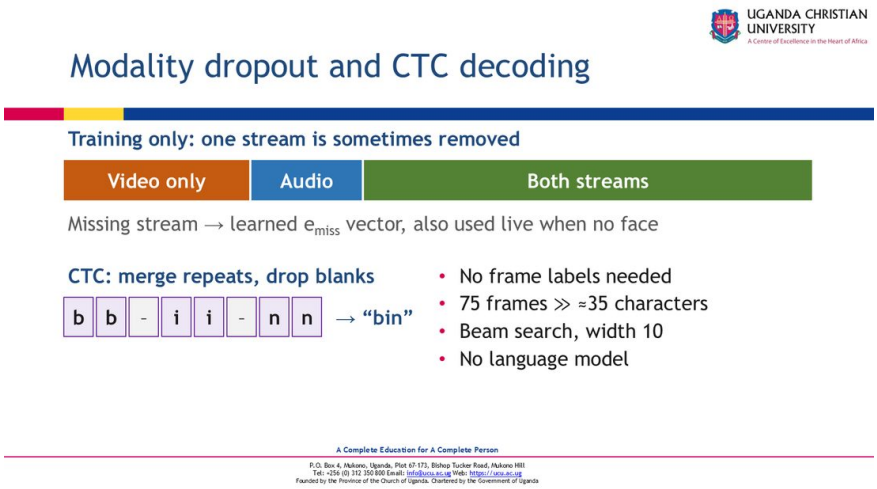
What is the difference between self-attention and cross-modal attention?

In cross-modal attention, queries come from one modality and keys and values from the other.

What if the gate does not change with noise?

That is a finding we would report, and we would diagnose it using the checklist in the evaluation plan.

Slide 9 | Modality dropout and CTC decoding (0:22)



WHAT TO SAY

- In training we sometimes remove one stream, so the model cannot ignore video. A missing stream becomes a learned vector, also used live when no face is found.
- CTC needs no frame labels: repeats merge and blanks drop. We decode with beam search and no language model. Nehum will explain evaluation.

BACKGROUND YOU SHOULD UNDERSTAND

A dropped stream is replaced by a learned 'missing modality' vector, not a mask; masking every key would make the attention softmax produce NaN. At inference, if no face is found in over half a window, the video is replaced by the same vector, giving clean audio-only fallback.

CTC [11] sums over every alignment of T frames to the text. Loss = -log p(text | input), PyTorch CTCLoss with blank 0 and zero_infinity. Condition: frames >= characters + repeats; 75 frames versus at most about 35 characters, asserted per item. Greedy decoding is used for validation; prefix beam search for final results and the demo.

QUESTIONS YOU MAY BE ASKED

Why no language model?

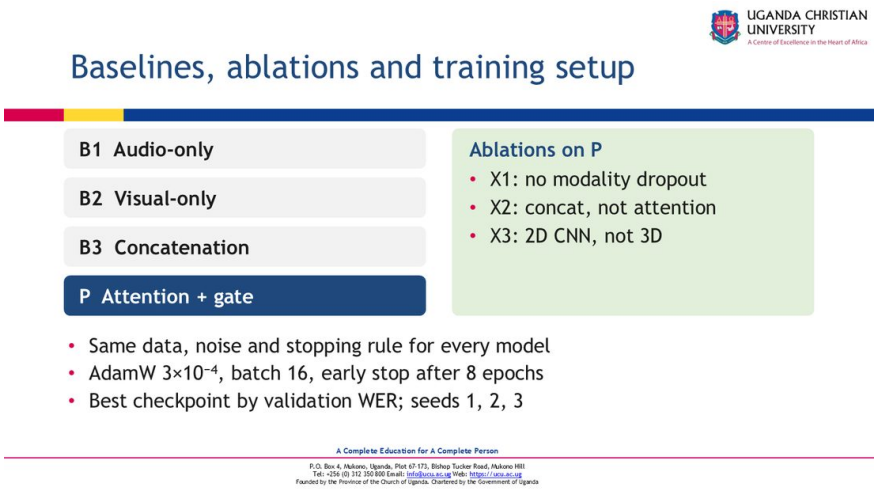
GRID's fixed grammar would correct errors by construction, inflating accuracy in a way that would not carry over to real captioning. A grammar decoder may be reported separately, clearly labelled.

Part 5: Baselines and evaluation

Presenter: **Ansasire Nehum Naamaye**

Slides 10–11 | suggested total 0:44

Slide 10 | Baselines, ablations and training setup (0:22)



WHAT TO SAY

- We compare our model P with three baselines: audio only, video only, and simple concatenation. Ablations remove modality dropout, attention, or the 3D convolution.
- Every model uses the same data, noise and training rules, so each comparison isolates one change.

BACKGROUND YOU SHOULD UNDERSTAND

Training (train.py): one YAML config; seeds set; config and commit hash saved; AdamW, peak learning rate 3e-4, weight decay 1e-2, 1-epoch warm-up then cosine decay; batch 16 with mixed precision; gradient clipping at 5; dropout 0.3 / 0.1; up to 60 epochs, early stopping after 8 without improvement in mean validation WER; best checkpoint restored for evaluation.

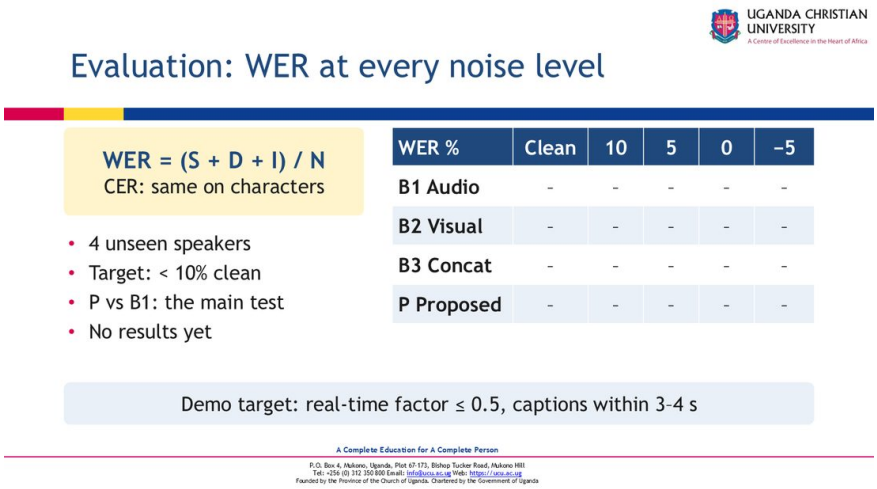
The audio-only baseline also trains on noisy audio; one trained only on clean audio would be unfairly weak.

QUESTIONS YOU MAY BE ASKED

What if fusion never beats audio-only?

We report it honestly with the diagnosis. B3 is still a working fused system, and the per-SNR analysis still shows where video helps or not.

Slide 11 | Evaluation: WER at every noise level (0:22)



WHAT TO SAY

- We report word and character error rate at every noise level on four unseen speakers. Our target is under 10% clean, with the biggest gains at 0 and minus 5 dB.
- The table is empty because we have no results yet. Isabelle will present the demo and plan.

BACKGROUND YOU SHOULD UNDERSTAND

Example: reference 'bin blue at f two now', hypothesis 'bin blue at s two now': one substitution in six words, WER 1/6, about 16.7%. WER is computed at corpus level with jiwer. CER is more sensitive on the letter slot.

Key comparisons: P vs B1 is the main test; P vs B2 at -5 dB (fused should be no worse than lips alone); P vs B3 isolates attention and dropout. If P is worse than B1 we check: did B2 learn; do gate values vary with SNR; overfitting to training faces; dropout too high; video misaligned.

Latency targets for the demo: real-time factor at most 0.5, caption latency about 3-4 s, median and 95th percentile over 100+ windows. Tests run with pytest before every merge, including an overfit test on 8 clips and a 5-frame video-shift sanity test.

QUESTIONS YOU MAY BE ASKED

Why not average across noise levels?

Averaging would hide exactly what we measure: how quickly accuracy degrades and where video starts to help.

How does your 10% target compare with published numbers?

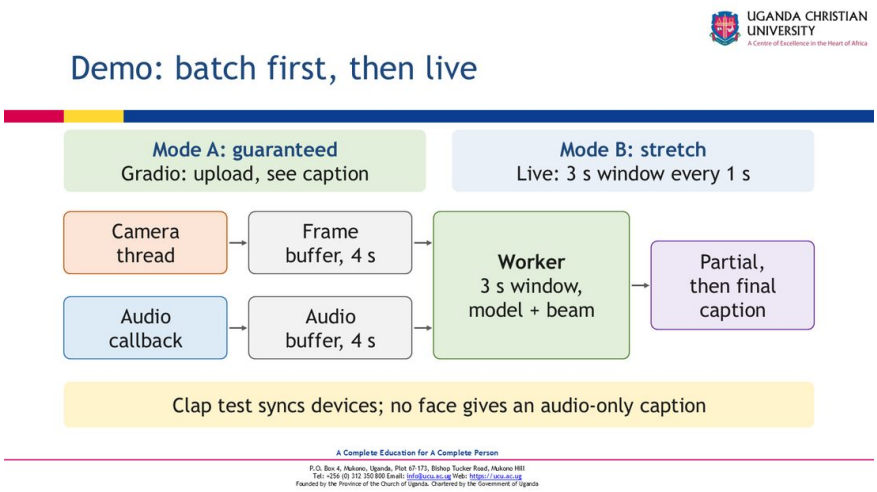
Published numbers are on other datasets such as LRS2, so they are not comparable. 10% is our project target on GRID.

Part 6: Demo, plan and risks

Presenter: **Wakisa Isabelle**

Slides 12–16 | suggested total 0:54

Slide 12 | Demo: batch first, then live (0:22)



WHAT TO SAY

- Mode A, our guaranteed demo, is a Gradio page: upload a clip, see the caption. Mode B captions live: camera and microphone fill buffers, and every second a worker decodes the last three seconds.
- A clap test syncs the devices, and if no face is found the caption falls back to audio only.

BACKGROUND YOU SHOULD UNDERSTAND

Live mode: a camera thread and an audio callback fill 4-second ring buffers with timestamps; for each of 75 instants the nearest frame is chosen; a hand-clap test removes fixed audio-video offset; partial captions update every second and a final caption appears after 500 ms of silence. If too slow: 1.5 s hop, smaller beam, GPU, quantisation.

QUESTIONS YOU MAY BE ASKED

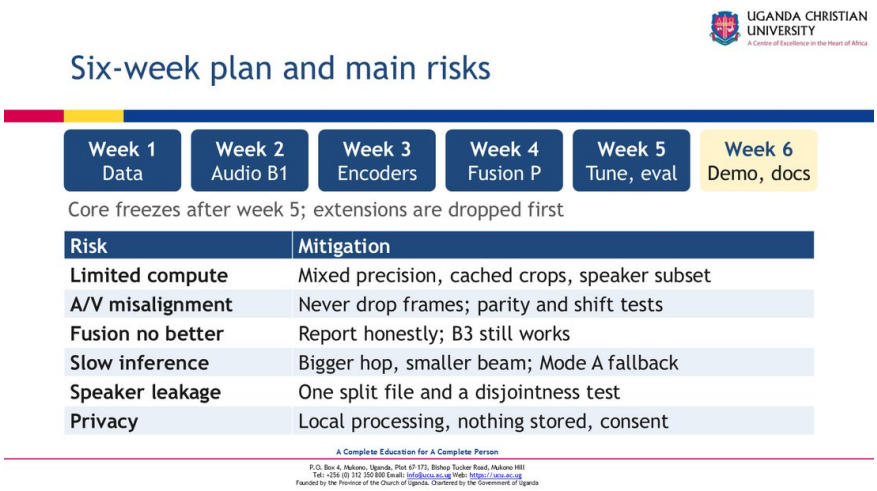
Is Mode B real-time?

Near real-time: captions appear about one window behind the speech, because the bidirectional encoders need future context. We measure and report the actual latency.

Will it work on normal conversation?

Not reliably. It is trained on GRID's fixed grammar, so the demo uses GRID-style sentences and we say so openly.

Slide 13 | Six-week plan and main risks (0:22)



WHAT TO SAY

- Six weeks, in dependency order, with the core frozen after week five.
- Main risks: limited compute, misalignment, fusion not helping, latency and privacy. Each has a mitigation, and if fusion does not help, we will report that honestly.

BACKGROUND YOU SHOULD UNDERSTAND

Milestones M1-M9 each have completion criteria, for example: M3, the audio baseline overfits 8 clips and is evaluated at all five SNRs; M8, Mode A matches the evaluation script; M9, a fresh clone reproduces one evaluation.

If the schedule slips, drop in this order: extensions, optional experiments X4-X6, Mode B, multiple seeds. Never dropped: baselines, the proposed model, per-SNR evaluation, ablations X1-X3. Work is split into a data pair, a model pair and an application pair.

Other risks: audio-video misalignment (never drop frames; parity and shift tests), CTC instability (warm-up, clipping), speaker leakage (one split file and a test), GRID differs from live speech (demo on GRID-style sentences).

Deliverables: the lipscribe package, data scripts, configs, train/evaluate/transcribe scripts, checkpoints, logs and plots, tests, the demos, README and the final report.

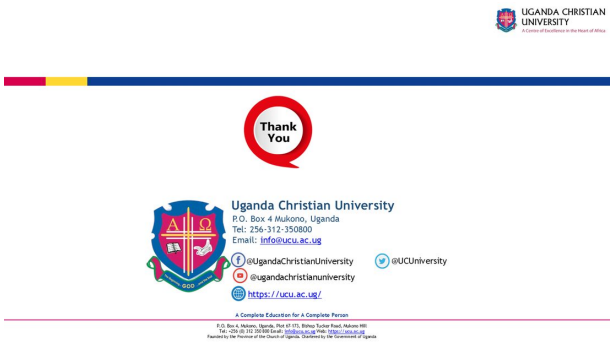
Closing line if time allows: we will report outcomes faithfully, including any where fusion does not help.

QUESTIONS YOU MAY BE ASKED

How is privacy handled?

Processing is local, nothing is stored by default, an indicator shows when capturing, and any recording needs written consent and is deleted after the project.

Slide 14 | Thank you (0:10)



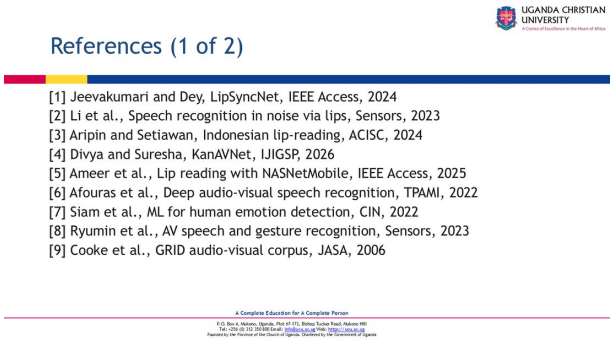
WHAT TO SAY

- Thank you. We are happy to take questions.

BACKGROUND YOU SHOULD UNDERSTAND

Questions: the presenter of the relevant slide answers first. Any member may be asked about any part, so everyone should read all six sections. The full 61-slide deck can be used as backup to show detail during questions.

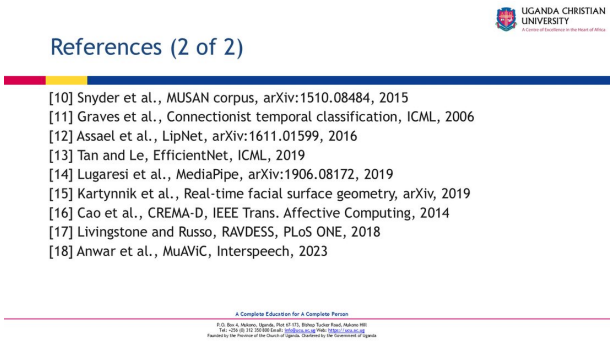
Slide 15 | References (1 of 2) (backup)



WHAT TO SAY

- Not presented. Shown only if asked.

Slide 16 | References (2 of 2) (backup)



WHAT TO SAY

- Not presented. Shown only if asked.